A Complete Classification of Pythagorean Triples via Generalized Fibonacci Sequences and Pisano Periods

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Abstract

We present a novel classification of all primitive Pythagorean triples into five disjoint families based on generalized Fibonacci sequences and their periodic behavior under modulo 9 arithmetic (Pisano periods). We prove that these five families—which we name Fibonacci, Lucas, Phibonacci, Tribonacci, and Ninbonacci—form a complete partition of the space of all Pythagorean triples when characterized by the digital root properties of their generating parameters. The families exhibit distinct Pisano periods of 24, 24, 24, 8, and 1 respectively, creating a 72-8-1 orbital structure. We further establish an *Inradius Identity*: for every BEDA tuple (b, e, d, a), the inradius of the associated right triangle equals be. As a corollary, all BEDA triangles with a prescribed inradius r are recovered by integer factorization of r alone, replacing Barning–Berggren tree search with a direct arithmetic lookup. Applications include number theory, dynamical systems, and cryptography.

1 Introduction

1.1 Background on Pythagorean Triples

A *Pythagorean triple* is an ordered triple of positive integers (C, F, G) satisfying the Pythagorean theorem:

$$C^2 + F^2 = G^2 \tag{1}$$

The classical parametric representation of primitive Pythagorean triples, known since Euclid, is given by:

$$C = m^2 - n^2, \quad F = 2mn, \quad G = m^2 + n^2 \tag{2}$$

where $m > n > 0$, $\gcd(m, n) = 1$, and $m \not\equiv n \pmod 2$.

While this parametrization is well-established, it does not reveal deeper structural patterns in the distribution of Pythagorean triples. In this paper, we introduce an alternative generating framework based on generalized Fibonacci sequences that reveals a hidden five-fold symmetry.

1.2 The BEDA Framework

We introduce the *BEDA tuple* representation, defined by four parameters (b, e, d, a) satisfying:

$$d = b + e \tag{3}$$
$$a = b + 2e \tag{4}$$

From any BEDA tuple, we generate a Pythagorean triple via:

$$C = 2de \tag{5}$$
$$F = ab \tag{6}$$
$$G = e^2 + d^2 \tag{7}$$

Proposition 1. *Every BEDA tuple (b, e, d, a) satisfying equations (3)–(4) generates a valid Pythagorean triple via equations (5)–(7).*

Proof. We verify $C^2 + F^2 = G^2$:

$$\begin{aligned} C^2 + F^2 &= (2de)^2 + (ab)^2 \\ &= 4d^2e^2 + a^2b^2 \\ &= 4(b+e)^2e^2 + (b+2e)^2b^2 \\ &= 4(b^2e^2 + 2be^3 + e^4) + (b^4 + 4b^3e + 4b^2e^2) \\ &= 4b^2e^2 + 8be^3 + 4e^4 + b^4 + 4b^3e + 4b^2e^2 \\ &= b^4 + 4b^3e + 8b^2e^2 + 8be^3 + 4e^4 \end{aligned}$$

Meanwhile:

$$\begin{aligned} G^2 &= (e^2 + d^2)^2 = (e^2 + (b+e)^2)^2 \\ &= (e^2 + b^2 + 2be + e^2)^2 = (b^2 + 2be + 2e^2)^2 \\ &= b^4 + 4b^3e + 8b^2e^2 + 8be^3 + 4e^4 \end{aligned}$$

Therefore $C^2 + F^2 = G^2$. □

1.3 Digital Roots and Pisano Periods

The *digital root* of a positive integer n, denoted $\mathrm{dr}(n)$, is defined as:

$$\mathrm{dr}(n) = \begin{cases} 9 & \text{if } n \equiv 0 \pmod 9 \\ n \bmod 9 & \text{otherwise} \end{cases} \tag{8}$$

The digital root maps integers to the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

For a Fibonacci-like sequence $\{F_n\}$ defined by the recurrence $F_n = F_{n-1} + F_{n-2}$ with initial conditions $F_0 = a, F_1 = b$, the *Pisano period* $\pi(m)$ is the period with which the sequence repeats when taken modulo m. In this work, we focus specifically on $m = 9$.

2 The Five Families

Definition 1 (The Five Families)**.** *We define five generalized Fibonacci sequences, each with initial terms* (F_0, F_1) *and recurrence relation* $F_n = F_{n-1} + F_{n-2}$*:*

1. ***Fibonacci Family****:* $(F_0, F_1) = (1, 1)$
2. ***Lucas Family****:* $(F_0, F_1) = (2, 1)$
3. ***Phibonacci Family****:* $(F_0, F_1) = (3, 1)$
4. ***Tribonacci Family****:* $(F_0, F_1) = (3, 3)$
5. ***Ninbonacci Family****:* $(F_0, F_1) = (9, 9)$

Each family generates a sequence of (b, e) pairs via $(b, e) = (F_n, F_{n+1})$.

2.1 Pisano Periods of the Five Families

Theorem 2 (Pisano Periods under Modulo 9)**.** *The five families exhibit the following Pisano periods when their digital roots are computed modulo 9:*

Family	***Initial Terms***	***Pisano Period*** $\pi(9)$
Fibonacci	$(1, 1)$	*24*
Lucas	$(2, 1)$	*24*
Phibonacci	$(3, 1)$	*24*
Tribonacci	$(3, 3)$	*8*
Ninbonacci	$(9, 9)$	*1*

Proof. The Pisano periods are established through direct computation. For each sequence, we compute the digital roots of consecutive terms and detect when the pattern first repeats. The computations are provided in Appendix A, with full verification available in the accompanying code repository.

For example, the Fibonacci sequence modulo 9 produces the digital root cycle:

$$\{1, 1, 2, 3, 5, 8, 4, 3, 7, 1, 8, 9, 8, 8, 7, 6, 4, 1, 5, 6, 2, 8, 1, 9\}$$

which repeats after 24 terms. □

3 Main Result: Complete Partition

Theorem 3 (Complete Partition of Digital Root Pairs)**.** *The five families partition the set of all possible digital root pairs* $\{1, 2, \ldots, 9\}^2$ *into five disjoint subsets. Specifically:*

1. *The Fibonacci family contributes 24 unique* $(dr(b), dr(e))$ *pairs*
2. *The Lucas family contributes 24 unique* $(dr(b), dr(e))$ *pairs*

3. *The Phibonacci family contributes 24 unique* $(dr(b), dr(e))$ *pairs*

4. *The Tribonacci family contributes 8 unique* $(dr(b), dr(e))$ *pairs*

5. *The Ninbonacci family contributes 1 unique* $(dr(b), dr(e))$ *pair*

The total is $24 + 24 + 24 + 8 + 1 = 81 = 9^2$*, accounting for all possible digital root pairs with* no overlaps.

Proof. We establish this result computationally by:

1. Computing the Pisano cycles for each of the five families

2. Extracting the unique $(\mathrm{dr}(F_n), \mathrm{dr}(F_{n+1}))$ pairs from each cycle

3. Verifying that the union of these five sets contains exactly 81 pairs

4. Verifying that the intersection of any two distinct sets is empty

The complete classification table is provided in Table 1. □

Table 1: Complete Classification of Digital Root Pairs by Family

dr(b)\dr(e)	1	2	3	4	5	6	7	8	9
1	F	F	L	P	F	P	L	F	F
2	L	L	F	L	P	P	L	F	L
3	P	P	T	L	F	T	F	L	T
4	F	P	F	P	P	L	L	P	P
5	P	L	L	P	P	F	P	F	P
6	L	F	T	F	L	T	P	P	T
7	F	L	P	P	L	F	L	L	L
8	F	L	P	F	P	L	F	F	F
9	F	L	T	P	P	T	L	F	N

F = Fibonacci, L = Lucas, P = Phibonacci, T = Tribonacci, N = Ninbonacci

4 Implications and Orbital Structure

4.1 The 72-8-1 Structure

The five families naturally group into three *orbital classes* based on their Pisano periods:

- **24-Cycle Orbit ("Cosmos")**: Fibonacci + Lucas + Phibonacci = $24+24+24 = 72$ pairs
- **8-Cycle Orbit ("Satellite")**: Tribonacci = 8 pairs
- **1-Cycle Orbit ("Singularity")**: Ninbonacci = 1 pair (the fixed point $(9, 9)$)

This 72-8-1 structure has been independently verified through computational experiments with Markovian dynamical systems on the BEDA parameter space.

We now give a group-theoretic proof that this structure is not coincidental.

4.2 Group-Theoretic Origin: Five Families as Orbits of the Fibonacci Matrix

Let $\mathbb{Z}/9\mathbb{Z}$ denote the integers modulo 9, with elements $\{0, 1, \ldots, 8\}$ (where 0 corresponds to digital root 9). The digital root map $\mathrm{dr} : \mathbb{Z}_{>0} \to \{1, \ldots, 9\}$ induces a well-defined map $(\mathrm{dr}(b), \mathrm{dr}(e)) \in (\mathbb{Z}/9\mathbb{Z})^2$.

Define the *Fibonacci matrix*

$$F = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z}/9\mathbb{Z}),$$

which acts on $(\mathbb{Z}/9\mathbb{Z})^2$ by $(b, e) \mapsto (e,\, b + e)$. This is exactly the Fibonacci recurrence step.

Theorem 4 (Group-Theoretic Classification)**.** *The five families of Theorem 3 are exactly the orbits of the cyclic group $\langle F \rangle$ acting on $(\mathbb{Z}/9\mathbb{Z})^2$. Specifically:*

1. *F has order 24 in $\mathrm{GL}_2(\mathbb{Z}/9\mathbb{Z})$, with $\det F = -1$ and $F^{24} = I$.*
2. *The action produces exactly 5 orbits with sizes 24, 24, 24, 8, 1, matching the five families.*
3. *The unique fixed point $(0, 0)$ is the Ninbonacci family; the length-8 orbit is the Tribonacci family; the three length-24 orbits are the Fibonacci, Lucas, and Phibonacci families.*
4. *The three cosmos orbits are distinguished by the $Q(\sqrt{5})$ norm $N(b, e) = b^2 + be - e^2$ reduced mod 9: since $\det F = -1$ the action maps $N \mapsto -N$, so each cosmos orbit carries a norm pair $\{N, -N\}$ with values*

$$\textit{Fibonacci} : \{1, 8\}, \quad \textit{Lucas} : \{4, 5\}, \quad \textit{Phibonacci} : \{2, 7\}.$$

5. *The Tribonacci orbit satisfies $N(b, e) \equiv 0 \pmod 9$ for all its elements; its stabilizer is $\langle F^8 \rangle \cong \mathbb{Z}/3\mathbb{Z}$.*

Proof. All claims are verified by exhaustive computation over the 81 elements of $(\mathbb{Z}/9\mathbb{Z})^2$, which we describe explicitly.

Order of F. Direct matrix exponentiation mod 9 gives $F^{24} = I$ and $F^k \neq I$ for $k < 24$; note $\det F = -1 \equiv 8 \pmod 9$, so $F \notin \mathrm{SL}_2(\mathbb{Z}/9\mathbb{Z})$ but $F^2 \in \mathrm{SL}_2(\mathbb{Z}/9\mathbb{Z})$.

Orbit computation. Starting from each $(b, e) \in (\mathbb{Z}/9\mathbb{Z})^2$ and iterating F until the orbit closes, one obtains exactly 5 distinct orbits with sizes $1 + 8 + 24 + 24 + 24 = 81$.

Matching to families. The initial conditions $(b_0, e_0) = (1, 1), (2, 1), (3, 1), (3, 3), (0, 0)$ (in $\mathbb{Z}/9\mathbb{Z}$ coordinates) belong to distinct orbits. Cross-checking each orbit against the classification of Theorem 3 confirms that every element of a given orbit carries the same family label, and distinct orbits carry distinct labels.

Norm invariant. For the Fibonacci step $F : (b, e) \mapsto (e, b + e)$,

$$N(e, b + e) = e^2 + e(b + e) - (b + e)^2 = -(b^2 + be - e^2) = -N(b, e).$$

So F negates the norm. On a length-24 orbit $F^{24} = I$ multiplies the norm by $(-1)^{24} = 1$, consistent. The norm values $\{N, -N \pmod 9\}$ are thus constant on each orbit, and computation gives the three pairs stated. $\square$

Remark 1. *The order of F in $\mathrm{GL}_2(\mathbb{Z}/9\mathbb{Z})$ equals the Pisano period $\pi(9) = 24$: both measure the period of the Fibonacci sequence modulo 9. Theorem 4 thus explains* why $\pi(9) = 24$ *is the "right" period for the five-family classification — it is the order of the Fibonacci generator acting on the digital-root state space. The 72-8-1 decomposition is the orbit-type decomposition of this cyclic group action.*

4.3 Connection to Pythagorean Triples

Corollary 5. *Every Pythagorean triple (C, F, G) generated from a BEDA tuple (b, e, d, a) can be uniquely classified into one of the five families based on $(dr(b), dr(e))$.*

This provides a new perspective on the structure of Pythagorean triples, complementary to the classical (m, n) parametrization.

4.4 Inradius as a Navigation Coordinate

The BEDA parametrization exposes a striking invariant: the inradius of every generated right triangle equals the product be.

Theorem 6 (Inradius Identity)**.** *For any BEDA tuple (b, e, d, a) satisfying* (3)–(4)*, the associated Pythagorean right triangle (C, F, G) has inradius*

$$r = b \cdot e.$$

Proof. The inradius of a right triangle with legs C, F and hypotenuse G is

$$r = \frac{C + F - G}{2}.$$

Expanding each side using $d = b + e$ and $a = b + 2e$:

$$\begin{aligned} C &= 2de = 2e(b + e) = 2be + 2e^2, \\ F &= ab = (b + 2e)b = b^2 + 2be, \\ G &= e^2 + d^2 = e^2 + (b + e)^2 = b^2 + 2be + 2e^2. \end{aligned}$$

Therefore

$$C + F - G = (2be + 2e^2) + (b^2 + 2be) - (b^2 + 2be + 2e^2) = 2be,$$

and $r = 2be/2 = be$. □

Corollary 7 (Inradius Navigation)**.** *Given a positive integer r, the complete set of BEDA-generated Pythagorean right triangles with inradius r is in bijection with the ordered factor pairs (b, e) of r with $b, e > 0$. For each such pair, the full tuple and triple are recovered deterministically as*

$$d = b + e, \quad a = b + 2e, \quad C = 2de, \quad F = ab, \quad G = e^2 + d^2.$$

Proof. Theorem 6 shows $r = be$ is necessary. Conversely, every positive factorization $r = be$ produces a valid BEDA tuple (both d and a are positive integers), which by Proposition 1 generates a valid Pythagorean triple. The map $(b, e) \mapsto (C, F, G)$ is injective since distinct ordered pairs (b, e) yield distinct pairs $(b^2+2be, 2be+2e^2)$ for F and C respectively. □

Remark 2 (Connection to the Barning–Berggren Tree)**.** *The Barning–Berggren ternary tree [5, 6] generates every primitive Pythagorean triple exactly once via three unimodular* 3×3 *matrices acting on* (C, F, G)*. Theorem 6 shows that this tree is simultaneously traversing the* divisor lattice *of* $r = be$*: each tree node corresponds to a pair* (b, e)*, and the three child transforms correspond to divisor-lattice moves. The BEDA coordinate system thus provides a flat, arithmetic parameterization of what the Barning–Berggren tree explores hierarchically. In particular, a tree search for triples with a given inradius* r *reduces to integer factorization of* r*—a substantial algorithmic simplification.*

Example 1. *Take* $r = 6$*. The factor pairs are* $(1, 6), (2, 3), (3, 2), (6, 1)$*, giving four BEDA seeds:*

(b, e)	d	a	C	F	G
$(1, 6)$	*7*	*13*	*84*	*13*	*85*
$(2, 3)$	*5*	*8*	*30*	*16*	*34*
$(3, 2)$	*5*	*7*	*20*	*21*	*29*
$(6, 1)$	*7*	*8*	*14*	*48*	*50*

Each triple is verified to satisfy $C^2 + F^2 = G^2$ *and to have inradius 6.*

5 Examples

We provide examples of Pythagorean triples from each family:

5.1 Fibonacci Family

From $(b, e) = (1, 1)$: $(C, F, G) = (4, 3, 5)$
From $(b, e) = (2, 3)$: $(C, F, G) = (30, 16, 34)$
From $(b, e) = (5, 8)$: $(C, F, G) = (208, 105, 233)$

5.2 Lucas Family

From $(b, e) = (2, 1)$: $(C, F, G) = (6, 8, 10)$
From $(b, e) = (3, 4)$: $(C, F, G) = (56, 33, 65)$
From $(b, e) = (7, 11)$: $(C, F, G) = (396, 203, 445)$

5.3 Tribonacci Family (8-Cycle)

From $(b, e) = (3, 3)$: $(C, F, G) = (36, 27, 45)$
From $(b, e) = (6, 9)$: $(C, F, G) = (270, 144, 306)$
From $(b, e) = (15, 24)$: $(C, F, G) = (1872, 945, 2097)$

5.4 Phibonacci Family

From $(b, e) = (3, 1)$: $(C, F, G) = (8, 15, 17)$
From $(b, e) = (4, 5)$: $(C, F, G) = (90, 56, 106)$
From $(b, e) = (9, 14)$: $(C, F, G) = (644, 333, 725)$

5.5 Ninbonacci Family (Fixed Point)

From $(b, e) = (9, 9)$: $(C, F, G) = (324, 243, 405)$
All multiples: $(648, 486, 810)$, $(972, 729, 1215)$, etc.

6 Applications

6.1 Cryptography

The orbital structure of the five families has applications in cryptographic key exchange. The difficulty of determining which orbit a given digital root pair belongs to (the *orbit decomposition problem*) can form the basis of a novel asymmetric encryption scheme, which we term BEDA cryptography.

6.2 Computational Number Theory

The classification provides an efficient algorithm for:

- Generating Pythagorean triples by family
- Classifying existing triples into families
- Predicting properties of large Pythagorean triples via digital root analysis

6.3 Dynamical Systems

The 72-8-1 orbital structure arises naturally in discrete dynamical systems governed by modular arithmetic. This connects the static classification of Pythagorean triples to the dynamics of recurrence relations.

7 Conclusion and Future Work

We have established a complete classification of Pythagorean triples into five families based on generalized Fibonacci sequences and their Pisano periods under modulo 9 arithmetic. This classification reveals a 72-8-1 orbital structure with applications in number theory, cryptography, and dynamical systems.

Future research directions include:

- Extending the classification to higher moduli ($m > 9$)

- Investigating connections to elliptic curves and modular forms
- Developing efficient algorithms for the orbit decomposition problem
- Exploring applications in lattice-based cryptography
- Characterizing which factor pairs (b, e) of a given inradius r lie in each of the five families; by Theorem 6 and Theorem 3, this is equivalent to classifying the divisors of r by their digital-root pair $(\mathrm{dr}(b), \mathrm{dr}(e))$
- Exploiting Corollary 7 to replace Barning–Berggren tree search with direct inradius factorization in computational number-theory applications

Acknowledgments

[To be added]

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A Computational Verification

All results in this paper have been computationally verified using Python 3.x with NumPy. The complete source code, including:

- Generators for all five families
- Pisano period computation
- Partition verification

- Pythagorean triple generation and validation

is available at: [GitHub repository URL to be added]